Intra Projects CPP Module 09 Edit

5 – 6 minutes

Prerequisites

The code must compile with c++ and the flags -Wall -Wextra -Werror

Don't forget this project has to follow the C++98 standard. Thus,

C++11 (and later) are NOT expected.

The purpose of this module is to use the STL. Then, using the

containers is authorized.

Any of these means you must not grade the exercise in question:

A function is implemented in a header file (except for template functions).

A Makefile compiles without the required flags and/or another compiler than c++.

Any of these means that you must flag the project with "Forbidden

Function":

Use of a "C" function (*alloc, *printf, free).

Use of a function not allowed in the exercise guidelines.

Use of an external library, or features from versions other than C++98.

Yes No

Code review

Check that a makefile is present with the usual compilation rules.

Check in the code that the program uses at least one container.

The person being evaluated must explain why they chose to use this

container and not another. >>> std::map (clé, value) >>> (date, valeur du bitcoin).

If not, the evaluation stops here. >>

Yes No

Error handle

You must be able to use an empty file or a file with errors. How to do that ? BitcoinExcahnge (we have the date et numbers of bitcoin.

(a basic example exists in the subject). The program must not stop

its execution before having performed the operations on the whole

file passed as argument.

You can use a wrong date. Wrong date.

You can enter a value greater than 1000 or less than 0. (errors)

If there is any problem during the execution then the evaluation stops here.

Yes No

Main usage

You must now use the "input.csv" file located at the top of this page.

You can modify this file with the values you want.

You have to run the program with the input.csv file as parameter.

Please compare some dates manually with the specified value.

If the date does not exist in the database, the program will have to use the

nearest lower date.

Yes No

2nd exercice.

Code review

Check that a makefile is present with the usual compilation rules.

Check in the code that the program uses at least one container.

The person being evaluated must explain why they chose to

use this container and not another ? >>> here we use std::stack >>> parce que on a besoin de. Selon le principe de FILO ?

If not, the evaluation stops here.

If the container chosen here is present in the first exercise

then the evaluation stops here.

Yes No

Main usage

Check that the program runs correctly using different formulas of your choice.

The program is not required to handle expressions with parenthesis or decimals

number.

If there is any problem during the execution then the evaluation stops here.

Yes No

Usage advanced

Check that the program runs correctly using different formulas of your choice.

Here is some tests:

8 9 * 9 - 9 - 9 - 4 - 1 +

> Result: 42

9 8 * 4 * 4 / 2 + 9 - 8 - 8 - 1 - 6 -

> Result: 42

1 2 * 2 / 2 + 5 * 6 - 1 3 * - 4 5 * * 8 /

> Result: 15

You can use the examples in the topic if you don't know which formula to use.

If there is any problem during the execution then the evaluation stops here.

3 exercice

Yes No

Code review

Check that a makefile is included with the usual compilation rules.

Check in the code that the program uses at least two containers.

If not, the evaluation stops here.

The person being evaluated must explain why they chose to use these containers

and not another?

Check in the code that the merge-insert sort algorithm is present and is used for each container. The Ford-Johnson algorithm must be used.

If the Ford-Johnson sorting algorithm is not used, the evaluation stops here and no points are awarded for any subsequent sections.

The student must be capable of explaining the following concepts:

The key aspects of merge insertion, specifically the role of pairs.

The Jacobsthal sequence and its relevance.

The process of binary search.

A brief explanation is expected. In case of doubt, the evaluation stops here and no points will be awarded for the subsequent parts of the evaluation.

If one of the containers chosen here is included in one of the previous

exercises then the evaluation stops here.

Yes No

Main usage

If the Ford-Johnson algorithm is used and all code review criteria are satisfied, you can now manually check that the program works correctly by using between

5 and 10 different positive integers of your choice as program arguments.

If this first test works and gives a sorted sequence of numbers you can continue.

If not, the evaluation stops now.

Now you have to check this operation by using the following command as an

argument to the program:

For linux:

`shuf -r -i 1-1000 -n 3000 | tr "\n" " " `

For OSX:

`jot -r 3000 1 1000 | tr '\n' ' '`

If the command works correctly, the person being evaluated should be able to
explain the difference in time used for each container selected.

If there are any problems during the execution and/or explanation then the
evaluation stops here.

Note: No points should be awarded for this section if the Ford-Johnson algorithm is not used, as per the Code Review section.

Yes No